

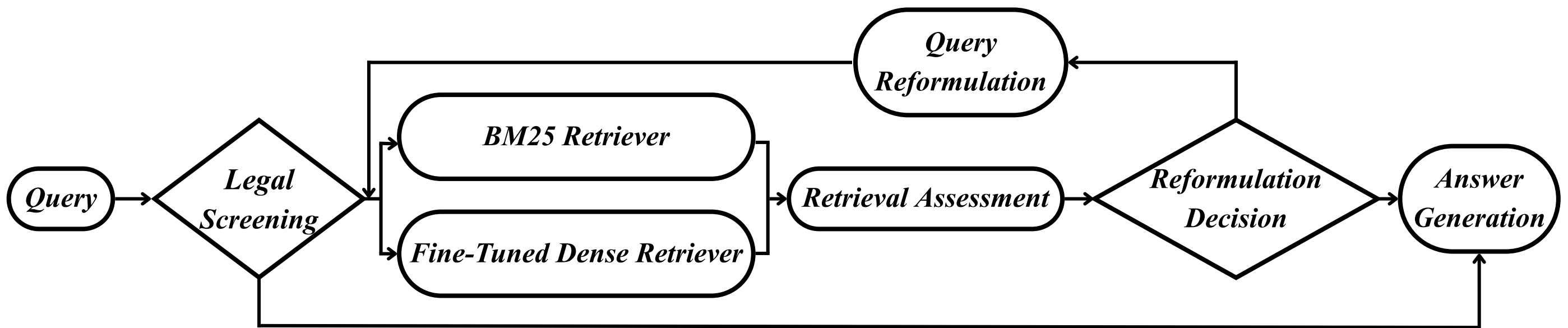
Legal Case Retrieval: An RAG-Based Legal Assistant

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Abstract

Traditional legal case retrieval relies heavily on keyword matching, which often struggles with semantic variation and complex legal queries. This project proposes a *retrieval-augmented legal question-answering system* that combines *BM25*, a *fine-tuned dense retriever*, and an *agentic multi-step workflow* for query reformulation and evidence assessment. Experimental results show improvements in *correctness* and *faithfulness* over both LLM-only and baseline RAG settings. The project demonstrates how hybrid retrieval and evidence-grounded generation can improve reliability in domain-specific question answering.

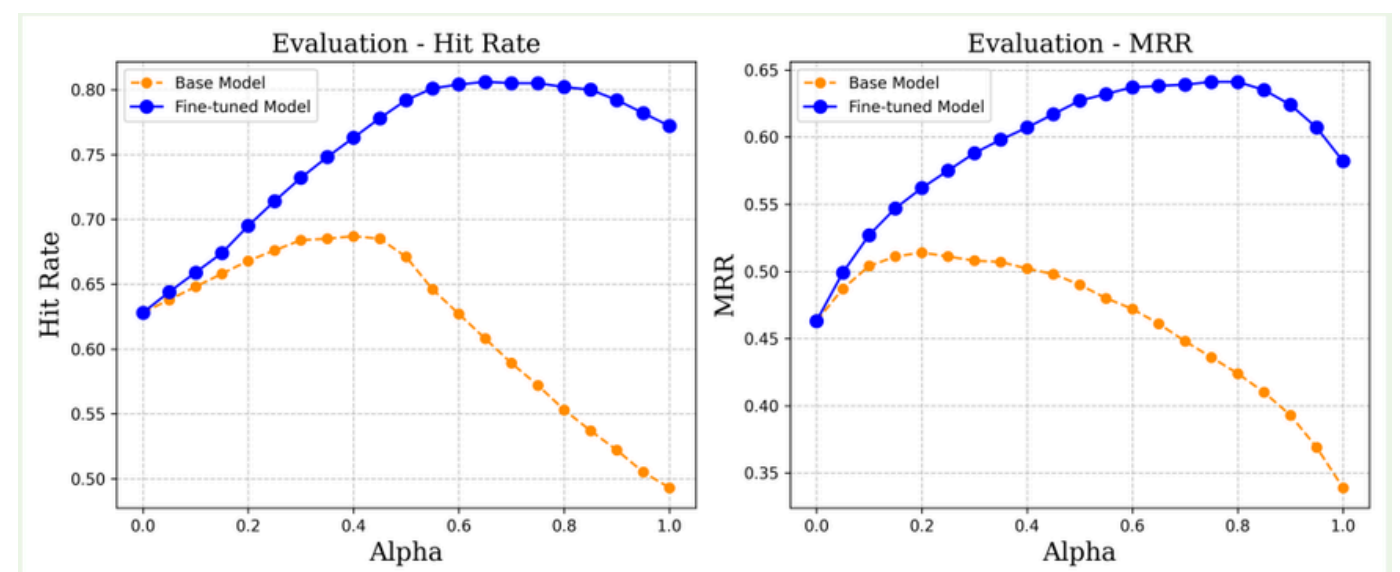
Architecture of the Proposed RAG Framework



Methodology

- Data Collection and Q&A Construction:** Judicial Yuan rulings were collected as the legal corpus. An LLM was used to generate a synthetic *Q&A dataset* for retriever fine-tuning and downstream evaluation.
- Dense Retriever Fine-Tuning:** A multilingual embedding model was fine-tuned on the Q&A pairs using *contrastive learning*, improving *HitRate@10* by *0.23* over the base model.
- Hybrid Retrieval and Agentic RAG:** The system combined *BM25* with the fine-tuned dense retriever in *Weaviate*. Built with *LangGraph*, the workflow assessed retrieval adequacy, reformulated queries when necessary, and generated answers from accumulated evidence.
- Evaluation:** System performance was assessed using *correctness* and *faithfulness*. The proposed agentic RAG framework achieved the strongest overall performance among the compared settings.

Results and Evaluation



System Configuration	Correctness	Faithfulness
LLM-Only	0.86	N/A
Naïve RAG (Baseline)	0.96	0.89
Agentic RAG (Proposed)	1.00	0.99

Alignment with the SDGs:

SUSTAINABLE
DEVELOPMENT GOALS



Technical Takeaways

This project demonstrates my ability to design and evaluate a domain-specific retrieval system by integrating *information retrieval*, *representation learning*, and *evidence-grounded generation*, with an emphasis on rigorous evaluation, reliable outputs, and data-driven decision support in domain-sensitive settings.